

Physical resolution of the Planck PR4 (NPIPE) anomaly by WRM

Recent analyses of Planck PR4 (NPIPE) have improved the accuracy of the standard model, but at the same time, they have fixed "deviations that should be corrected" that cannot be explained by Λ CDM. These can be mathematically resolved by the structural variables of WRM.

1. Slight increase in Hubble constant (H_0) and fixation of tension

In PR4 analysis $H_0 \approx 67.4$. The preceding and succeeding values were determined with extremely high confidence, proving that the deviation from the nearest observation (73.0) is not due to statistical measurement error.

WRM's answer:

This **67.4** is the base frequency (effective expansion rate) of the universe, calculated backward from the geometrically corrected "effective resonance radius of approximately 67 billion light-years" (46.5 billion light-years). The deviation from the neighboring region is the geometric offset. $\alpha = 0.046$. This can be formulated as an "extension of the optical path" resulting from the refractive index gradient of spacetime.

2. Anomaly of the Cosmic Lens Effect (A_L)

The Planck data exhibits an unnatural bias, where the gravitational lensing effect appears approximately 10% stronger than predicted. $A_L > 1$ remains.

WRM's answer:

This is not gravity caused by matter (dark matter), but rather a lensing effect caused by the **"refractive index gradient of spacetime itself" resulting from interference with external waves**. $\alpha = 0.046$. The non-uniform spatiotemporal density distribution along the axis adds a systematic excess (residual) to the lens calculations of the standard model.

3. Deficiency of low-order multipoles (large-scale structures)

In PR4, the phenomenon of fluctuation amplitudes being significantly lower than theoretical predictions at the largest scales in the universe (lower-order modes, especially quadrupoles) continues to be observed.

WRM's answer:

This is direct evidence that the universe is not infinitely vast, but rather a resonant system confined to a **physical outer shell of one trillion light-years**. In a region with an effective radius of 67 billion light-years, standing wave interference with the outer shell causes wavelengths close to the outer shell scale (lower-order modes) to cancel out (node) through destructive interference.

WRM Integrated Correction Mathematics

The "unexplained fractions" shown in the latest PR4 data converge to a single physical causality through the following structure constants.

$$R_{effective} \approx R_{geometric} \times \frac{1}{\sqrt{1 - \alpha^2}} \Big|_{\alpha=0.046} \approx 67.0 \text{ Gly}$$

(※ Compared to the geometric 46.5 Gly, $\alpha = 0.046$ (This determines the physical "effective size")

Conclusion:

The limitations of modern astronomy revealed by Planck PR4 suggest that the universe is a **"finite resonating box with spacetime refraction."** WRM achieves complete consistency with the latest observational data by making the **"residuals"** that the Standard Model discarded the **"core"** of its theory.